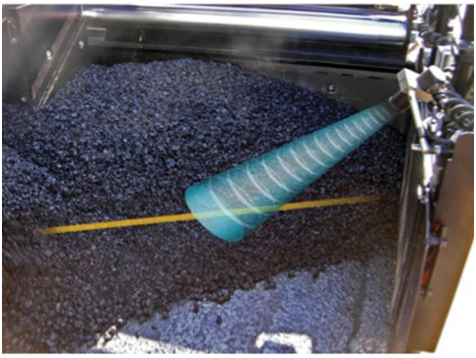


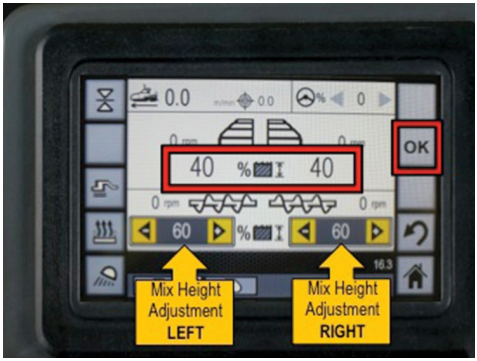
STEP 11: Set Feeder Sensor Position *(Continued)*

Sonic feeder sensors should be set perpendicular to the material flow 46 cm (18 in) away from the material. Always target the moving material.



STEP 12: Set Feeder Controls

On the left side of the tractor control panel touch screen, select the material feed adjustment icon.



STEP 12: Set Feeder Controls *(Continued)*

Set the conveyor control dials to 40% for each conveyor. Using the arrows on the touchscreen, adjust the mix height to 60%. After adjustments are made to the mix height push the OK icon on the touchpad to accept the setting. This is a starting point for the system. The operator or screed operator can fine tune these when paving begins.



STEP 13: Fill The Auger Chamber

Now fill the auger chamber in front of the screed until the material is covering one half the auger.

Fill the chamber with the engine at low idle, one side at a time, alternating between the conveyor and auger. This helps prevent material from being pulled under the machine and overfilling the chambers.

Using the manual override switches, alternately convey and auger material to the auger chamber and out to the ends of the augers.

The goal is to fill the auger chamber evenly on both sides.



STEP 13: Fill The Auger Chamber *(Continued)*

Use one conveyor switch at a time to move material out until it just touches the auger shaft.

Then, use the manual auger switch to move material out to the end of the screed. The correct head of material covers one half of the augers.

Always fill the auger chamber using the manual mode at low idle and be careful not to overfill.

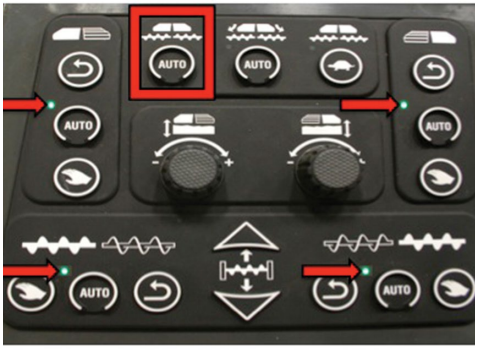
NOTE: Do not fill the chamber using the conveyor and auger switches together or by using the feeder override switches located on the screed. This will overfill the chamber and cause a bump on take off.



STEP 13: Fill The Auger Chamber *(Continued)*

Do not force material out to the end gate. You will overfill the chamber. Instead, move some material with a shovel to the end gate even with the face of the main screed.

NOTE: Do not fill in the area directly in front of the extender on rear-mounted extender screeds. This area will be filled by material automatically as the paver pulls forward off the starting reference.



STEP 13: Fill The Auger Chamber *(Continued)*

When you have filled the auger chamber half full—no more, no less—put the conveyor and auger mode switches on the operator's console to Auto. Either push the automatic conveyor and auger button or press individual automatic buttons for both the conveyor and auger. Verify that all four automatic system indicators are illuminated.



STEP 14: Set Accessory Functions

At this point it needs to be determined what accessory functions are going to be used on the machine.



STEP 14: Set Accessory Functions *(Continued)*

Automatic grade and slope controls are considered accessory controls. Set them according to to the job specifications. Install any averaging devices and controls as required.



STEP 14: Set Accessory Functions *(Continued)*

Step 14 is setting the accessory functions of the machine.

Refer to the machine Operation and Maintenance Manual for all the functions on the machine. Here are just a few:

- Automatic Engine Speed Control
- Friction Steer
- Vibrator System On/Off and Speed
- Screed Assist System
- Timers
- Propel Hill Hold



STEP 15: Pull Off Starting Reference

Make sure the gear selector is in the PAVE mode (1), the screed is in float (2), and the engine throttle is set at the chosen idle speed (ECO or high idle) (3). Then, turn the speed control dial (4) clockwise until the calculated paving speed is is displayed in the target speed area of the display (5), release brake (6). Propel the machine by squeezing the trigger and stroking the propel lever (7) full forward in one quick, smooth motion.

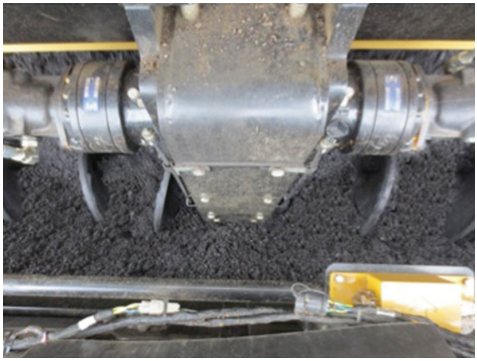
NOTE: When calculated target speed is reached, use the propel lever to stop and start the machine using the same quick, smooth motion as before. Slowly stroking the propel lever will result in a more uneven head of material and have an adverse effect on mat smoothness.



STEP 15: Pull Off Starting Reference *(Continued)*

Screed personnel observe the level of material at the outboard ends of the auger shafts to see if the auger shafts are one half covered.

There are individual material height dials on the screed control boxes which are adjusted to control the height of material at the outboard ends.

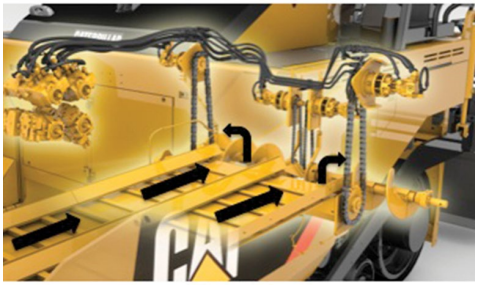


STEP 15: Pull Off Starting Reference *(Continued)*

As the paving operation stabilizes, the operator should check the speed of auger rotation. The augers should be turning uniformly in the range of 20-40 rpm.

To slow down the augers, the operator turns the ratio control dials clockwise. This action sends more mix to the auger chamber. To increase auger speed, turn the conveyor ratio control dials counterclockwise.

By all means, avoid erratic or ON/OFF operation of the augers. These conditions tend to create material segregation stripes in the mat and loss of smoothness.



STEP 15: Pull Off Starting Reference *(Continued)*

As the paving operation stabilizes, the operator should check the speed of auger rotation. The augers should be turning uniformly in the range of 20-40 rpm.

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By all means, avoid erratic or ON/OFF operation of the augers. These conditions tend to create material segregation stripes in the mat and loss of smoothness.

No Separation Marks In Mat
(Rear Mounted Extenders)



STEP 15: Pull Off Starting Reference *(Continued)*

As paving speed and feeder system operation are stabilized, screed personnel should look for transition marks between the main screed and screed extenders.

If screed height, angle of attack and slope are correct, the mat should have a uniform texture across the full width of the mat and there should be no longitudinal lines present.

Extender Screeds Too Low
(Rear Mounted Extenders)



STEP 15: Pull Off Starting Reference *(Continued)*

If an extender is set too low, there will be a longitudinal transition mark in line with the inner edge of the screed extender. The mat will be thinner behind the screed extender.

Use the extender height switch to raise the extender until no transition mark is present.

Extender Screeds Too High
(Rear Mounted Extenders)



STEP 15: Pull Off Starting Reference *(Continued)*

If an extender is set too high, there will be a longitudinal transition mark in line with the outer edge of the main screed. The mat will be thicker behind the screed extender.

Use the power height switch to lower the extender until no transition mark is present.



The most important fundamental of smooth paving is maintaining a consistent paving speed and constant head of material.

Normally, you can set up the paver to pave smoothly at any speed that matches the delivery of mix to the job site. **The important thing is to keep the speed constant on all systems.**

If the paving speed is changed drastically, the screed will either rise or fall and mat smoothness will suffer.

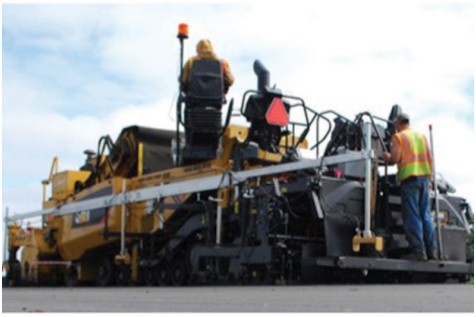
Also, if paving speed changes, the demands on the feeder system change. So the feeder system controls will have to be adjusted to match the new material demands. For quality paving results, always follow the basic fundamentals of paver set-up and keep the operation consistent.



The Cat Grade & Slope System for the F-Series Asphalt Pavers uses high accuracy sensors and software to control the screed's relative vertical position to a control reference during paving operations. The system can be configured for operation with one or more mechanical and sonic sensors.

On the F-series Asphalt Paver any of the four displays allow operators to monitor and adjust system parameters through a graphical user interface.

For more information regarding the Caterpillar Grade and Slope System for the F-Series Asphalt Pavers refer to the Operation & Maintenance Manual or the booklet Grade & Slope F-Series Pavers (Form #: QEDQ1959)



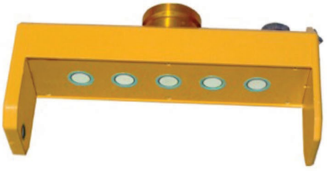
The Grade and Slope Electronic Control Module (ECM) receives operator inputs from the machine displays via a network of CAN bus and Ethernet communication lines. These inputs are analyzed, the ECM then works in conjunction with the machine systems to control the tow points.

The system has the “Auto Valve Calibration” feature which executes a calibration program that allows the system to determine the response of the valves to valve drive commands with no manual intervention.



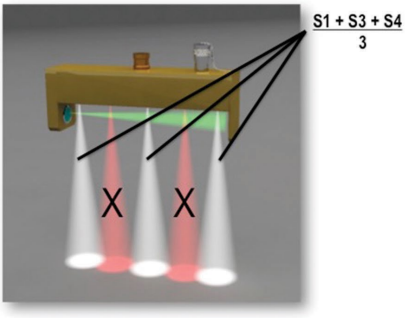
The control system is integrated into the displays on the tractor and on the screed. The operator no longer needs to install or remove the control system panels from the paver making it quicker and easier for the crew to prepare the machine at the beginning of the shift or at the end.

The sensors are the same as earlier Cat Grade and Slope so they can be used on the new system.



The ST200 is an elevation control sensor. The sonic tracer contains five sonic transducers and one temperature compensation transducer.

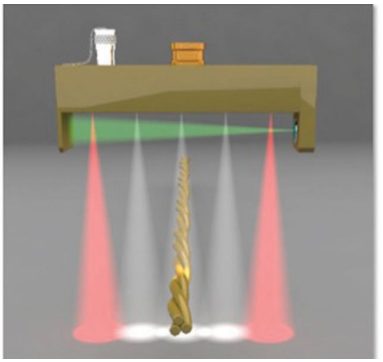
The transducers emit sound pulses that reflect off physical objects, such as a surface or a string line. As the sound pulses strike an object, a portion of the sound pulses reflect back to the transducers. By using the speed of sound, the sonic tracer calculates the exact distance to the control reference by using the elapsed time for the reflected pulse to return to the transducer. Once the sonic tracer has been nulled, any change in distance will be sent to the control system. The system evaluates the information, and sends a signal to the ECM to raise or lower the tow point cylinder.



The temperature sensor signal bounces off of the end plate on the sensor. This distance is fixed so any change in the speed of sound that is thermally induced is evaluated by the system software and changes are made, if needed, to the tow point position.

The working range for the sonic tracer is 20 cm (8.0 in) to 100 cm (39 in) and has an adjustable working window.

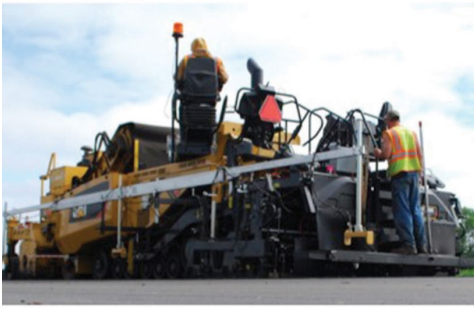
In surface mode, the five transducers average the surface of the reference object. The sonic tracer must be mounted parallel to the direction of travel with the connector to the rear. The sonic tracer must be configured for surface tracking by using the Sensor Select icon and choosing the correct icon on the display touch screen.



In string line mode, the three center transducers average the string line and the two outer transducers signal the display for horizontal guidance.

The sonic tracer must be mounted perpendicular to the direction of travel with the connector on the left when viewed from the rear. This enables the guidance arrows on the display to be directionally correct.

The sonic tracer must be configured for string line mode by using the Sensor Select icon and choosing the correct icon on the display touch screen.



An averaging ski is an elevation control device. The ski is mounted on swing arms attached to the tow arm. The ski removes small surface irregularities and averages out longer surface irregularities. You can use up to three ST200 sonic tracers on the averaging ski in the following combinations:

- Three sonic tracers spaced at equal distances from each other.
- Two sonic tracers with one mounted at each end of the beam.
- One sonic tracer mounted in the center of the beam.

The readings from each active sonic tracer on the ski are averaged in order to provide a smoother reference elevation.

The ski articulates to enable the rear sonic to be positioned over the finished mat.